

Duality Gap in Dual Convex Optimization of ReLU Networks

The Non-Convex Mystery, Solved?

A survey, 2020–2025 (Emphasis: Pilanci & Ergen)

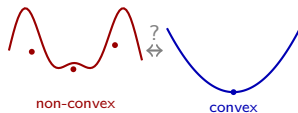
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April 8, 2026

The Mystery

- SGD was designed for convex problems
- We run it on provably non-convex losses
- It works anyway — why?
- The folk answer: *“overparameterization, good init, luck”*
- **A better answer exists. It starts in 2020.**



Duality Gap in 45 Seconds

Every optimization problem has a companion **dual**.

$$\begin{array}{ccc} P^* & \geq & D^* \\ \text{primal (hard problem)} & & \text{dual (companion)} \end{array}$$

$$\text{gap} = P^* - D^*$$

$$\text{strong duality} \iff \text{gap} = 0$$

For convex problems the gap is zero. For non-convex — usually not.

For non-convex problems this almost never holds... **almost**.

The Bridge Is Built: (2020)

Non-convex primal:

$$\min_{\{\mathbf{u}_j, \alpha_j\}_{j=1}^m} \frac{1}{2} \left\| \sum_{j=1}^m (\mathbf{X}\mathbf{u}_j)_+ \alpha_j - \mathbf{y} \right\|_2^2 + \frac{\beta}{2} \sum_{j=1}^m (\|\mathbf{u}_j\|_2^2 + \alpha_j^2) \quad (\text{primal})$$

$\Downarrow$ zero duality gap

Equivalent convex program (SOCP):

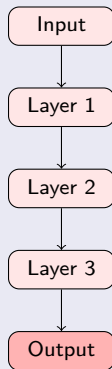
- $O((n/r)^r)$ variables; finite, exact
- Width condition: $m \geq m^* \leq n + 1$

Theorem 1 [, 2020]

Two-layer ReLU + weight decay is **secretly convex**. Strong duality holds for $m \geq m^*$.

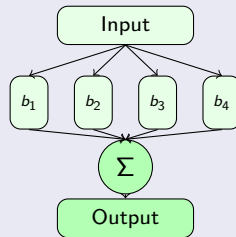
The Climax: Depth Breaks, Parallelism Restores

Standard L -layer network



$$P^* > D^* \text{ for } L \geq 3$$

Parallel (K branches) network

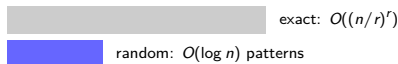


$$P^* = D^* \text{ for any } L$$

Architecture, not depth, is the villain.

The theory shipped.

Theory: (2024)



Approximation guarantee (Theorem 2.1):

$$P^* \leq \tilde{p}^* \leq C\sqrt{\log 2n} P^*$$

First poly-time approx. guarantee for regularized ReLU NNs.

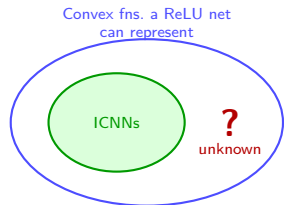
Practice: (2023)



- Matches/beats tuned Adam
- Essentially no hyperparameters
- Global opt. guarantee (2-layer)

Open Question: What Is a “Convex ReLU Network”?

- Training is convex. But which *networks* represent convex functions?
- Standard answer: **ICNNs** (Amos et al. 2017)
- (2025):
at depth ≥ 2 , $\text{ICNN} \subsetneq \text{ReLU-convex}$
- The right convex class is **still open**
- Matters for: optimal transport, energy-based models



Non-Stanford group: INRIA / Univ. de Lyon.

**ReLU network training was never really non-convex —
and whether the hidden convex program is exact, tractable,
and practically solvable depends on three things:**

width | architecture | $\log n$

width

Pilanci-Ergen 2020
 $m \geq m^*$

architecture

Wang-Ergen-Pilanci 2021
parallel branches

$\log n$

Kim-Pilanci 2024
 $\sqrt{\log n}$ approximation

Thank you — questions welcome.

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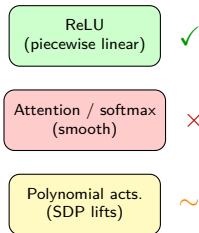
B1: Generalisation, Not Training Loss

- Zero duality gap = global optimum of **training loss**
- Test error / generalisation bounds: **NOT** covered
- Optimal-set polytope $\rightarrow$ hope for future generalisation bounds
- The most important honest thing to say about this work



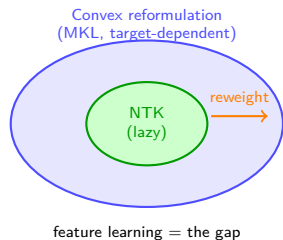
B2: Does This Extend to Transformers?

- Convex reformulation requires ReLU: piecewise linear
- Attention (softmax) is smooth — no hyperplane patterns
- Extension to transformers: **explicitly open** (, 2024 conclusion)
- Polynomial activations: partial via SDP (, 2021)



B3: Relationship to NTK

- NTK (Jacot 2018): infinite width, no feature learning
- Convex reformulation: exact at finite width, with feature learning
- (2023):
NTK is a special case of the MKL structure
- Iterative reweighting upgrades NTK to the full convex solution
- feature learning = the gap between NTK and full convex solution



B4: What About ResNets?

- ResNet residual block = 2-branch parallel network
- Branch 1: **identity** path
- Branch 2: **transform** path (Conv $\rightarrow$ BN $\rightarrow$ ReLU)
- (2021):
ResNets + weight decay $\subseteq$ parallel-network framework
- Strong duality restored: inherits zero-duality-gap result

